

FILTERSYNTHESIZER V1.0.3

User Manual

What every control does, which topology to choose, and what to do when the solver says no.

2026-09-22

Screenshots captured at v1.0.3

1. How the app is organized

The sidebar holds the specification. The five tabs turn it into hardware, left to right. Everything recomputes as you edit; there is no Run button.

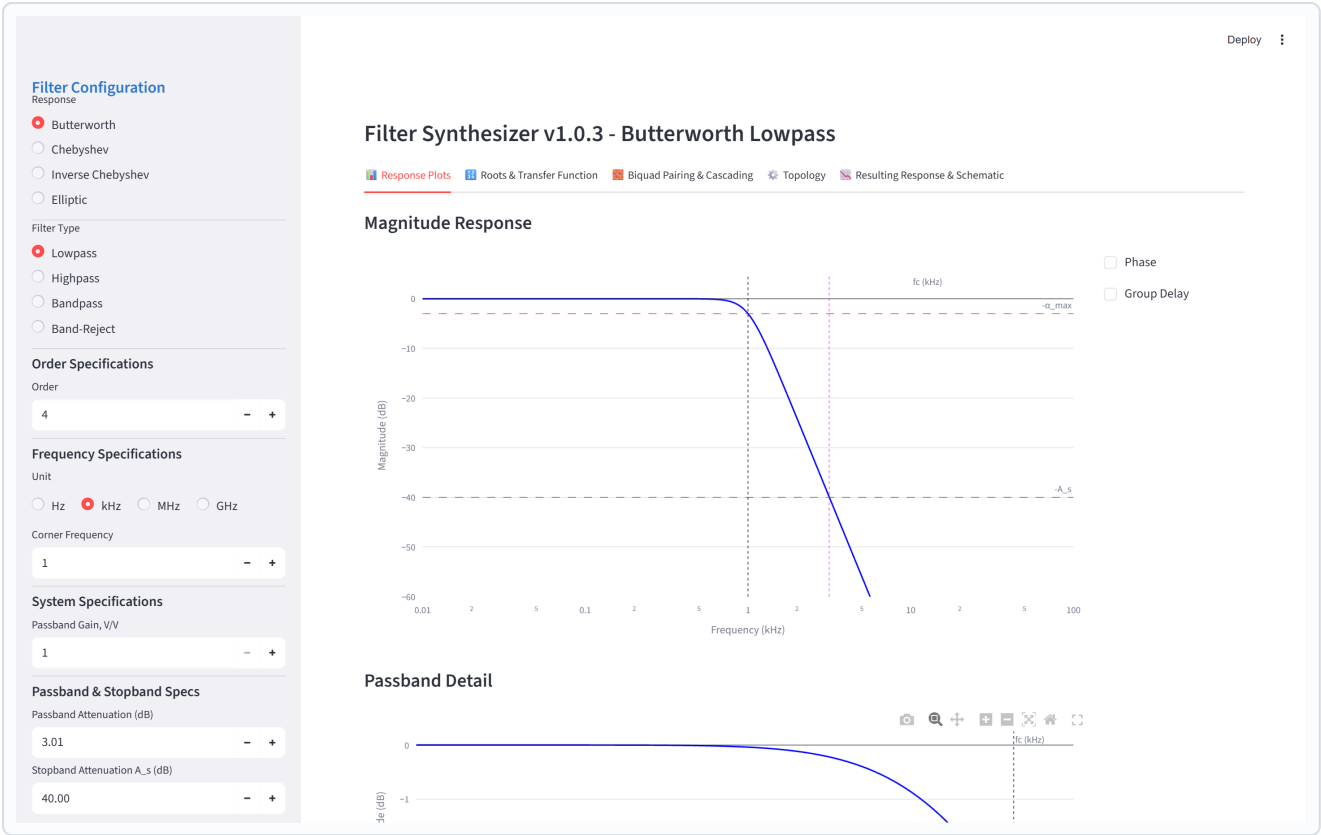


Figure 1. The app as it opens: the specification on the left, the five tabs across the top.

Tab	What it does	What it needs first
Response Plots	Magnitude (optionally phase and group delay) of the approximation; frequency probes; manual notches	A valid specification
Roots & Transfer Function	Poles, zeros, gain constant K, and $H(s)$ in three forms	—
Biquad Pairing & Cascading	Groups the roots into sections and shares out the gain; hands the cascade to the Topology tab	Every zero assigned to a section
Topology	Per section: circuit family, op-amp, component limits; solve , then pick one BOM	The cascade from Tab 3
Resulting Response & Schematic	The cascade built from your picks: Bode plot, Monte-Carlo, schematics, PDF report	A picked BOM in every section

Tabs 1–3 are mathematics and update at once, so that is where to iterate on order, corner and gain. Tab 4 is where time is spent: each section is solved on its own, and solving does not choose anything — you pick a row. Tab 5 and the report stay blocked until every section has a pick, and say which sections are still missing.

What persists and what resets

- **Nothing is saved to disk.** The design lives in the browser session. Reloading the page (F5) starts over: the sidebar returns to its defaults and every section has to be solved again. The PDF report is the record of a design.
- **The approximation recomputes a moment after each sidebar edit.** A specification you have used before comes back instantly from a cache.
- **Solving runs in the background.** Other sections and tabs stay usable while a section solves. Solves run one at a time: pressing **Solve section** on a second section queues it, and the tab shows how many are running.
- **The Topology tab refreshes itself every two seconds** while it is open — that is how finished solves appear without a click.
- **Results are remembered per configuration.** A section's results belong to everything that produced them: its f_0 , Q and gain, the envelope, the E-series, the op-amp and the convergence settings. Change any of these and the table disappears and the button reads **Solve section** again; change it back and the earlier table returns without a new solve.
- **A pick belongs to the section number.** It survives re-sorting the table and switching tabs. After changing the specification, solve and pick every section again — until you do, a section keeps its previous pick, and Tab 5 and the report use those parts.

2. Concepts and vocabulary

Term	Means
Section (stage)	One block of the cascade, realized by one circuit: a complex pole pair (2nd order), a pole pair with an absorbed real pole (3rd order), or a lone real pole (1st order). Zeros assigned to it make it a notch, high-pass or band-pass section. Sections are numbered by rising Q.
Absorption	An odd order leaves a real pole. Absorbing it into a pole-pair section makes that section 3rd order and saves an op-amp. Switched on in Tab 3; off by default.
f_o , Q, f_p , f_z	A section's pole-pair frequency and quality factor; f_p is its real pole (absorbed or lone); f_z its notch frequency. Shown in each section's header in Tab 4.
K_i	A section's gain constant — the leading coefficient of its transfer function, set by Tab 3's gain distribution. Tab 4 shows the passband gain it implies: $H(0)$ for a low-pass, $H(\infty)$ for a high-pass, $ H(f_o) $ for a band-pass.
Cell	A named circuit that realizes a section, e.g. 3LP-gained or 2BP-MFB-QE . The name encodes order, response and variant (§7.3).
BOM (candidate)	One complete set of component values for a section. A solve returns many, ranked.
Envelope	The limits a BOM must respect: $C_{min}...C_{max}$, $R_{min}...R_{max}$, Max R ratio, and the E-series.
Snapping	Resistors are solved as continuous values, then moved to values of your resistor E-series. Capacitors are taken from their E-series from the start and are never snapped.
Sens (sensitivity score)	How strongly the section's transfer-function coefficients move when a component drifts by 1 %. Lower is better. A ranking figure, not a predicted tolerance — Monte-Carlo in Tab 5 is that.
Snap cost	How far the snapped parts leave the section from its target: a weighted sum of the fractional errors in gain, pole frequency, Q and notch depth. 0 is perfect, lower is better. Like Sens, it ranks; it is not a percentage.
Design vs realized	Design is the mathematical target (blue curves). Realized is what the snapped parts do with the chosen op-amp model (red curves).
Sign	Inverting cells contribute -1 to the cascade. An odd number of them inverts the output.

3. Control reference

One table per tab, top to bottom as on screen. The **Key** column is the control's internal name; quote it when reporting a problem. In keys, ***** stands for the section number (or a band or notch index). Controls without a key show "—".

3.1 Sidebar — the specification

The sidebar is divided into two main sections: **Filter Configuration** and **Frequency Specifications**.

Filter Configuration includes:

- Response:** Radio buttons for Butterworth (selected), Chebyshev, Inverse Chebyshev, and Elliptic.
- Filter Type:** Radio buttons for Lowpass (selected), Highpass, Bandpass, and Band-Reject.
- Order Specifications:** A numeric input field for Order, currently set to 5.

Frequency Specifications includes:

- Unit:** Radio buttons for Hz, kHz (selected), MHz, and GHz.
- Corner Frequency:** A numeric input field, currently set to 2.
- System Specifications:** A numeric input field for Passband Gain, V/V, currently set to 1.5.

Figure 2. The sidebar, top and bottom, with the Quick Start example: Butterworth low-pass, order 5, 2 kHz, 1.5 V/V.

Control	Key	Default	What it does
Response	—	Butterworth	The approximation: Butterworth (maximally flat), Chebyshev (passband ripple), Inverse Chebyshev (flat passband, notched stopband), Elliptic (ripple in both, steepest). Changes the order limits and which boxes appear below.
Filter Type	—	Lowpass	Lowpass, Highpass, Bandpass, Band-Reject. The band types take two corners and an order per side.
Order	<code>widget_sym_order</code>	4	Filter order. For the band types it is the order of each side, labelled Order (LP & HP) . Limits below.
Asymmetric	<code>is_asym_checkbox</code>	off	Band types only (not for a Chebyshev or Elliptic band-reject): separate orders for the two sides.
LP Order / HP Order	<code>widget_lp_order</code> , <code>widget_hp_order</code>	4 / 4	Shown when Asymmetric is ticked. LP is the upper-edge side, HP the lower-edge side.
Unit	—	kHz	Unit of every frequency box and of the Tab 1 probes.
Corner Frequency	<code>widget_fc</code>	1	Lowpass and Highpass: the passband edge.
Lower / Upper Passband Corner	<code>widget_f1</code> , <code>widget_f2</code>	1 / 2	Band types: the two passband edges.

Control	Key	Default	What it does
Passband Gain, V/V	widget_gain	1	Overall passband gain, at least 1. Tab 3 shares it out among the sections.
Passband Attenuation (dB)	—	3.0103	Butterworth and Inverse Chebyshev: attenuation at the corner. 0.01–12 dB.
Passband Ripple $\alpha_{\max}$ (dB)	—	1.0	Chebyshev and Elliptic: the passband ripple, in the same box.
Stopband Attenuation A_s (dB)	sym_as_val	40	Minimum stopband attenuation, 10 dB or more. For Inverse Chebyshev and Elliptic it shapes the design; for Butterworth and Chebyshev it only sets the reported stopband edge f_s and the green line on the plot. The key applies to the Inverse Chebyshev and Elliptic band types; elsewhere the box has none.
Lower Stopband A_{sl} / Upper Stopband A_{su} (dB)	as_sl_val , as_su_val	40 / 40	Inverse Chebyshev and Elliptic band types with Asymmetric ticked: one attenuation per stopband.
Response Modifications	—	off	Checkboxes that appear only where they apply — see below.

Order limits

Response	Lowpass / Highpass	Bandpass, per side	Band-Reject, per side	Asymmetric, per side
Butterworth	1–20	1–15	1–14	1–14
Chebyshev	1–20	1–20	1–16	1–10 (Bandpass only)
Inverse Chebyshev	1–20	1–20	1–20	1–10
Elliptic	2–15	2–15	2–15	1–10 (Bandpass only)

A value outside the new limits is clamped when you change the response.

Coupled behaviour worth knowing

- **The two corners push each other apart.** Raising the lower corner to or above the upper one moves the upper corner to twice the lower; lowering the upper corner to or below the lower one moves the lower corner to half the upper.
- **The corner carries over between filter types.** The Lowpass/Highpass corner and the lower band corner are the same stored value.
- **The orders stay in step until you edit HP.** The symmetric order and the LP order are one value; the HP order follows it until you change HP once, after which it keeps its own.
- **Asymmetric stopbands start equal.** Ticking **Asymmetric** copies the common A_s into both A_{sl} and A_{su} ; unticking copies A_{sl} back.

Response Modifications

Checkbox	Appears for	Effect
Passband Even Order Modification	Chebyshev and Elliptic, even order, all types except Bandpass	An even-order Chebyshev or Elliptic response normally starts in a ripple valley: the gain at DC (at HF for a high-pass) sits one full ripple below the passband peaks. The modification puts it on a peak instead.
Lower / Upper Passband Even Ord Mod	Asymmetric Chebyshev or Elliptic Band-Reject, per even-order side	The same, for each passband separately.
Stopband Rolloff	Inverse Chebyshev and Elliptic	Moves one pair of transmission zeros to infinity: one notch fewer, and the stopband keeps falling beyond the last notch.
Lower / Upper Stopband Rolloff	Asymmetric Inverse Chebyshev or Elliptic Bandpass	The same, for each stopband separately.
Coincident Stopband Notches	Inverse Chebyshev and Elliptic Band-Reject, total order divisible by 4	The zero pair the rolloff would send to infinity lands at the centre of the stopband instead, so two notches coincide there.

Validation

An invalid specification replaces the whole main area with a red **Mathematical Constraint Violation** box (§6.1). The one you can actually hit: an asymmetric band-reject needs an even total order (LP + HP). Asymmetric Elliptic Bandpass also warns, without blocking, when LP + HP exceeds 15.

3.2 Tab 1 — Response Plots

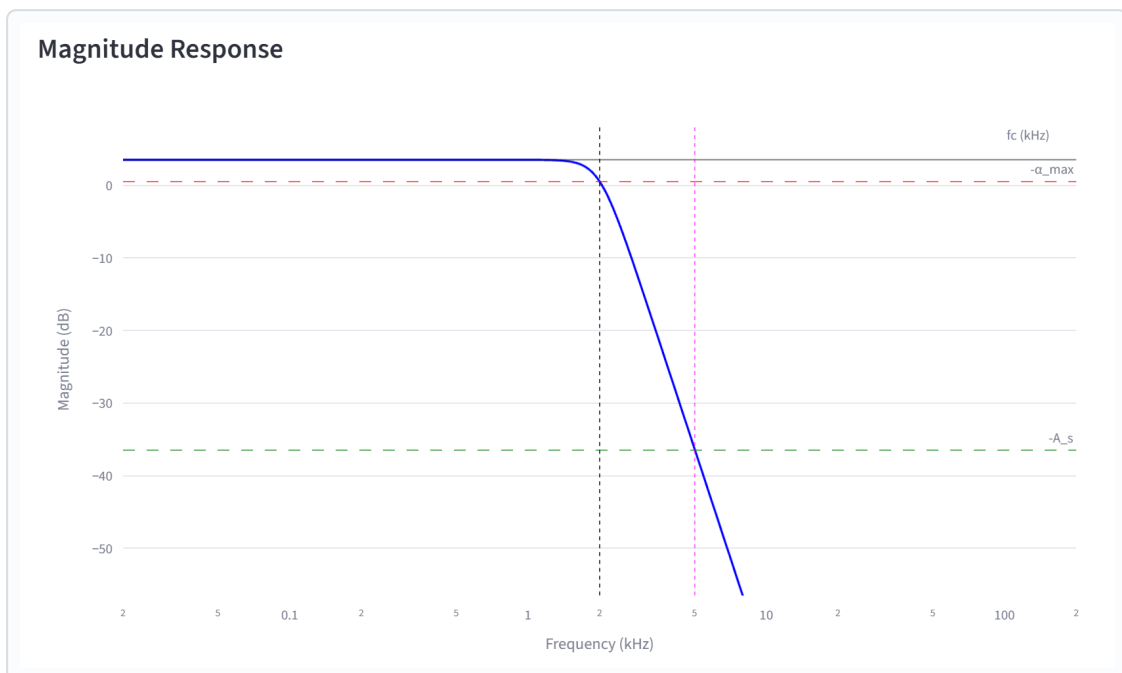


Figure 3. Magnitude response of the Quick Start example: flat at +3.52 dB (1.5 V/V), down to the red $-\alpha_{\max}$ line at the 2 kHz corner.

The **Magnitude Response** plot marks $-\alpha_{\max}$ (red dashed), $-A_s$ (green dashed), the corner (black dashed) and the calculated stopband edge f_s (magenta dotted). **Passband Detail** below zooms into the passband. Under the plots, the **Calculated Stopband Edge (f_s)** box is green when the stopband meets A_s , yellow when a pinned notch pushed the edge out, and red when the stopband never reaches A_s .

Control	Key	Default	What it does
Phase	—	off	Adds a Phase & Group Delay plot with the phase trace.
Group Delay	—	off	Adds the group-delay trace to the same plot.
Probe 1 / 2 / 3	probe_1_*, probe_2_*, probe_3_*	corner, 2× and 10× corner (Lowpass)	Gain and phase of the design at the typed frequency, printed under each box. Each filter type keeps its own probes.
Pin N	pin_notch_*	off	Manual notches, one row per zero pair the order allows. Butterworth and Chebyshev (Manual Notch Placement): an unpinned row reads ∞ ; pinning adds a finite notch at the typed frequency. Inverse Chebyshev and Elliptic (Manual Notch Tuning): unpinned rows show the computed notch; pinning freezes it so you can move it.
Notch frequency	val_notch_*	2× corner (Lowpass)	The pinned notch's frequency, in the sidebar unit.
Lower / Upper Stopband Notches	pin_notch_hp_*, val_notch_hp_*, pin_notch_lp_*, val_notch_lp_*	off	Bandpass: the same, one column per stopband.

Manual tuning is disabled for Elliptic band filters and for Elliptic Lowpass/Highpass above order 8; the tab says so.

3.3 Tab 2 — Roots & Transfer Function

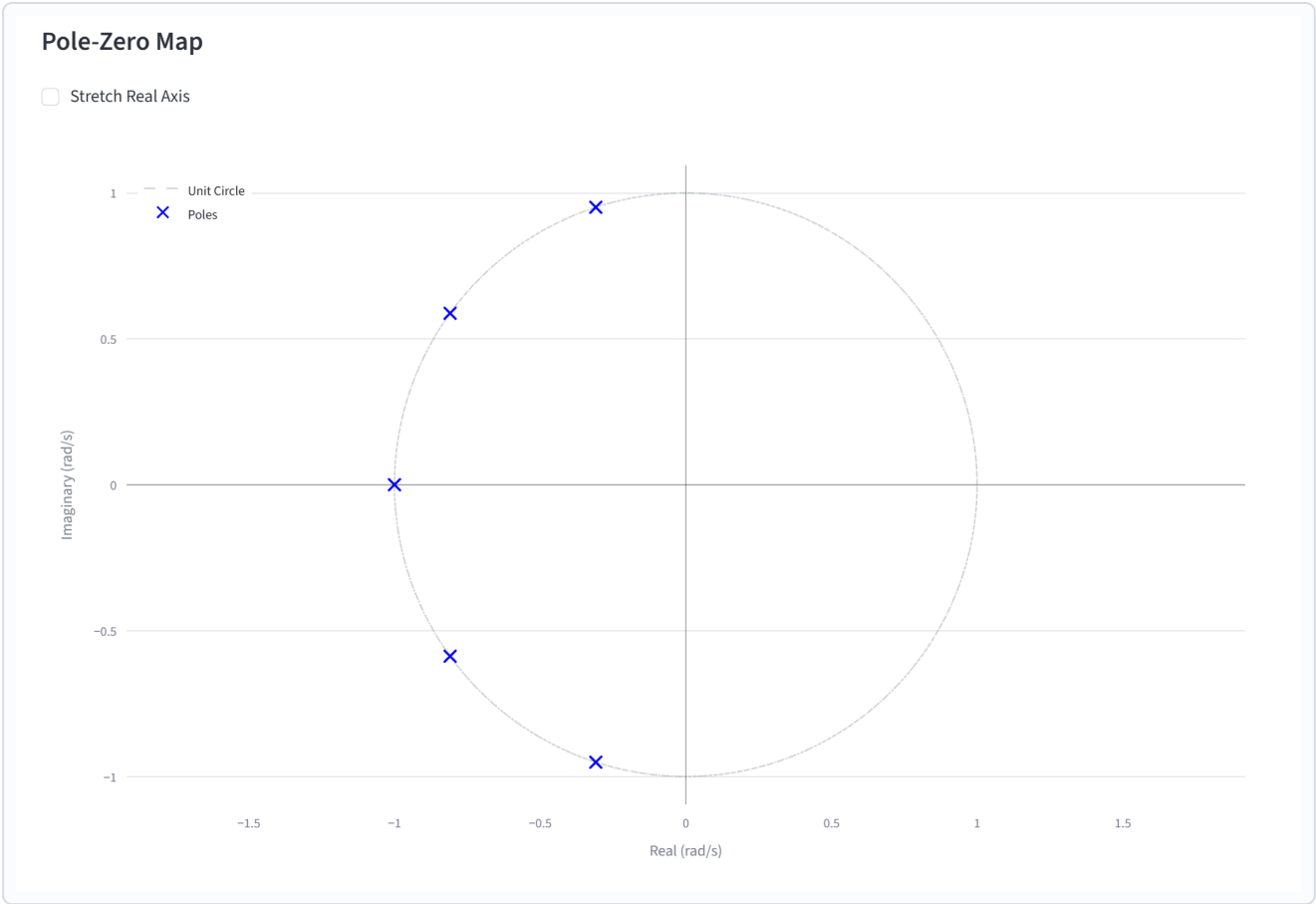


Figure 4. The pole-zero map, normalized: five poles on the unit circle — one real, two complex pairs.

Control	Key	Default	What it does
Domain Scale	<code>scale_roots</code>	Normalized	Normalized: corner (band centre for band types) at 1 rad/s, unity gain. Denormalized: real frequencies, and K includes the passband gain.
Pole-Zero Map Units	<code>unit_roots</code>	rad/s	Denormalized only: rad/s or Hertz for the map, the tables and H(s).
Stretch Real Axis	—	off	Stretches the map's real axis — useful when poles crowd the imaginary axis.
Magnification	—	1.0	Shown when stretching: 0.25–10.

Below the map: **Root Locations** (poles and zeros), the **System Gain Constant (K)** with its units, and **Transfer Function H(s)** in three expanders — Expanded Form (Isolated Gain Constant), Expanded Form (Distributed Gain Constant) and Factored Form (Cascaded Biquads). In each, the rendered equation wraps to fit the page; the grey box under it is the same expression as one LaTeX line, ready to copy, followed by the coefficient table.

3.4 Tab 3 — Biquad Pairing & Cascading

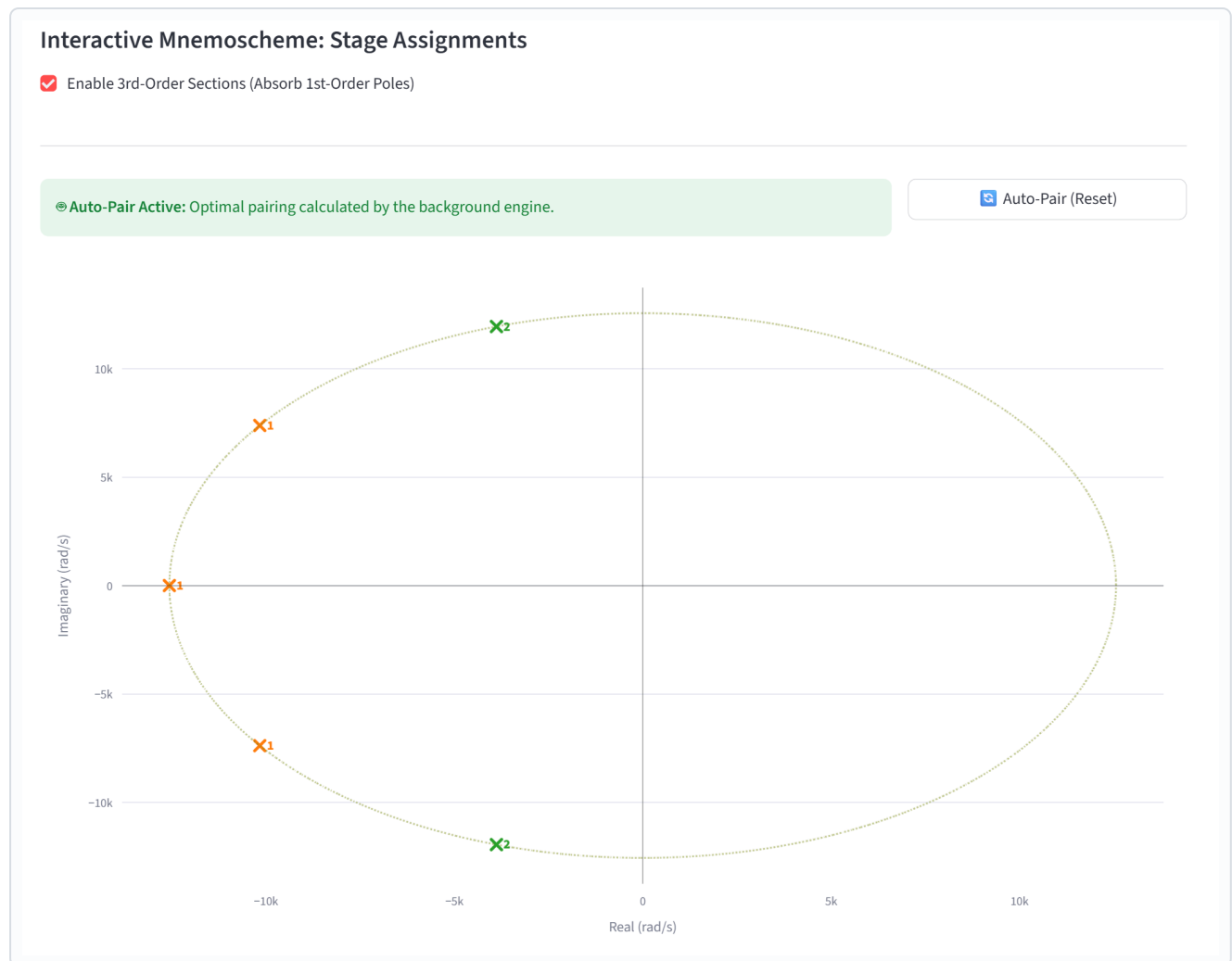


Figure 5. Tab 3 with absorption on: the real pole joins the low-Q pair as section 1 (orange); the high-Q pair is section 2 (green).

Control	Key	Default	What it does
Hardware Target Units	<code>unit_pair</code>	rad/s	Units of the mnemoscheme and of the section tables below. The Topology tab always works in Hz.
Enable 3rd-Order Sections (Absorb 1st-Order Poles)	—	off	Shown only when the filter has a real pole. Ticked: each real pole is absorbed into a pole-pair section, which becomes 3rd order. Unticked: the real pole is a 1st-order section of its own.
The mnemoscheme	<code>pz_mnemo_chart</code>	automatic pairing	The pole-zero map with each section in its own colour. Clicking edits the pairing (below).
 Auto-Pair (Reset)	—	—	Discards manual edits and restores the automatic pairing.
Remaining Gain Distribution	—	Distribute Remaining Gain Evenly	The pairing first sets every section to unity passband gain; the Calculated Remainder is the factor still needed to reach the specified passband gain (the gain over unity). This decides where it goes: shared evenly, all to the first stage, or all to the last stage. Band-Reject adds Equalize DC and HF gains of LP and HP sections , which makes both passbands land on the specified gain.

Editing the pairing by hand. Click a pole to select it, then click a zero to move that zero into the pole's section, or a real pole to absorb it (needs the checkbox), or the same pole again to cancel — or, on an absorbed real pole, to detach it into its own section. The banner turns to **Manual Override Active**; sections are renumbered by rising Q after every edit. Until every zero belongs to a section, the gain distribution and the section tables are withheld and the cascade is not handed on.

Each section below the map shows its transfer function (rendered and as a LaTeX line) and a table: K_i , ω_o or f_o , Q, the notch ω_z or f_z , the real pole ω_p where there is one, and **Peak Mag (V/V)** — the section's own maximum gain.

3.5 Tab 4 — Topology & Hardware Synthesis

Each section has a header line — order, fp for a real pole, f_o , Q, and f_z for a notch — then its settings expander, its gain control, the cell caption and the **Solve section** button. The caption names the cell the gain selects, e.g. *Cell: 3LP-gained · $H(0) = 1.5$ (from K) · tol ±2%*: a section gain within ±2 % of 1 uses the unity cell, above it the gained cell, below it an attenuating cell.

Convergence Settings

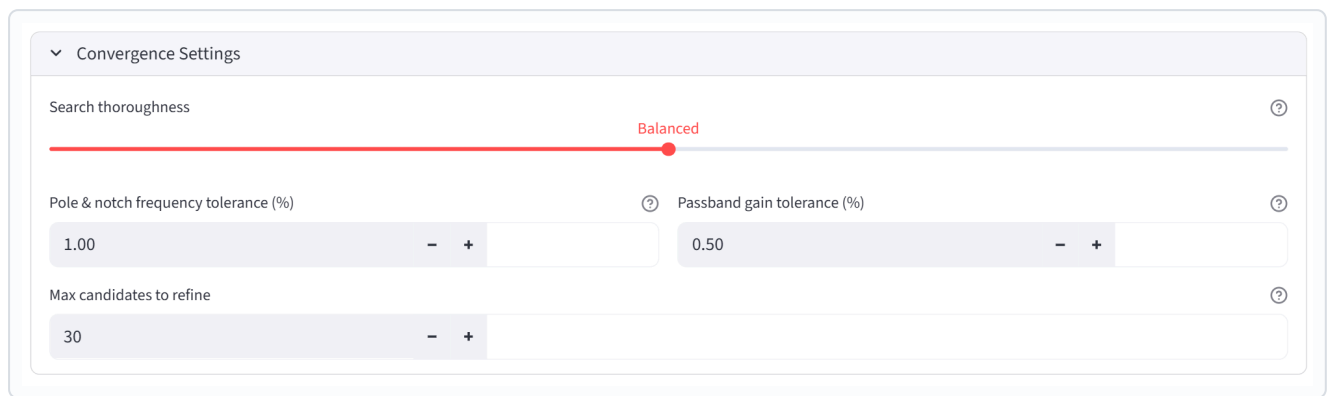


Figure 6. Convergence Settings. The defaults are right for a first run.

These apply to every section.

Control	Key	Default	What it does
Search thoroughness	hw_effort	Balanced	Fast / Balanced / Thorough: how many starting points the search tries and how many minima it keeps. Thorough finds more candidates and takes longer.
Pole & notch frequency tolerance (%)	hw_pole_tol_pct	1.00	How far a candidate's realized f_o — and a notch section's f_z — may sit from the target; used by the notch search path. Q is not gated by it.
Passband gain tolerance (%)	hw_gain_tol_pct	0.50	How far a candidate's realized passband gain may sit from the target: DC gain for low-pass and notch cells, HF gain for high-pass.
Max candidates to refine	hw_topk	30	Only the best N ideal solutions, by sensitivity, are corrected for the op-amp and snapped — the expensive step for gained and notch cells. Lower is much faster and lists fewer BOMs. 1–500.

Leave these alone until a solve returns nothing *and* the envelope is already wide; then raise **Search thoroughness**. Lower **Max candidates to refine** when a gained or notch section is slow. Loosening the tolerances admits candidates that miss the target by more — a last resort.

Per-section settings

Section 1 · order 3 · $f_p = 2,000\text{ Hz}$ · $f_0 = 2,000\text{ Hz}$ · $Q = 0.618$

Section 1 — topology & component settings

Topology family

☒ VCVS (Sallen-Key)
☐ MFB (Friend)
☐ AM (Ackerberg–Mossberg)

Op-amp model

Ideal (no op-amp limits)

Component envelope

C_min (μF)

6.80e-5

–

+

R_min (kΩ)

0.3000

–

+

Max R ratio

500.00

–

+

C_max (μF)

1.00e-2

–

+

R_max (kΩ)

2000.0

–

+

Capacitor E-series (single)

☐ E3
☐ E6
☒ E12

Resistor E-series (multiple)

☐ E12
☐ E24
☒ E48
☐ E96

Figure 7. Section settings: family, op-amp, component envelope and E-series, all at their defaults.

Control	Key	Default	What it does
Topology family	hw_fam_*	VCVS (Sallen-Key)	VCVS (Sallen-Key), MFB (Friend) or AM (Ackerberg–Mossberg). §4 helps choose.
Op-amp model	hw_opamp_choice_*	Ideal (no op-amp limits)	A library part or Custom.... With a real part the solver corrects for it and checks the BOM against its non-ideal model (§5).
A _{ol} (V/V), GBWP (Hz), R _o (Ω)	hw_aol_*, hw_gbwp_*, hw_ro_*	1e5, 1e6, 1200	Custom... only: open-loop gain, gain-bandwidth product, output resistance.
C _{min} , C _{max} (μF)	hw_cmin_*, hw_cmax_*	6.80e-05, 1.00e-02	Capacitor window: 68 pF to 10 nF.
R _{min} , R _{max} (kΩ)	hw_rmin_*, hw_rmax_*	0.3000, 2000.0	Resistor window: 300 Ω to 2 MΩ.
Max R ratio	hw_ratio_*	500	Largest over smallest resistor in one BOM. A wider spread is rejected. An MFB band-pass alone needs about 4·Q ² , so 500 allows Q up to about 11.
Capacitor E-series	hw_cser_*	E12	E3, E6 or E12 — one. Capacitor values come from this series directly. E3 is not usable yet (§6.3).

Control	Key	Default	What it does
Resistor E-series	hw_rser_*_*	E48	E12, E24, E48, E96 — any combination. Resistors are snapped to the union. With none ticked, E48 is used.
Equalize R, C values	hw_ameq_*	on	AM only. The handbook balanced design: $R_5 = R_6 = R_7 = R_8$ and $C_2 = C_3$. Untick to free R_5/R_6 and C_2/C_3 for a closer E-series fit; $R_7 = R_8$ stays matched.
Gained MFB	hw_mfbgain_*	off	MFB, 2nd-order high-pass-notch sections: solve the unity/gained realization instead of the attenuating one. Its HF gain lives in a narrow band, so Custom HF gain is switched on and pre-filled with the band's middle.
Lower Sensitivity MFB	hw_mfbbs_*	off	MFB, low-pass-notch sections: solve the low-sensitivity branch (Q set passively, $S_Q < 1$, a deeper null with a real op-amp).
Eliminate R1	hw_mfbbs_nor1_*	off	2nd-order low-pass notch with Lower Sensitivity MFB ticked: drop R_1 for a 7-part cell. The DC gain is then pinned to a narrow window, so Custom DC gain is switched on and pre-filled.

A 1st-order section has a shorter expander, ⚙ **Section N — component settings**: the op-amp, **C_max** (the pole is realized at the three E-series capacitor values nearest below it), and the two E-series. It needs no family and no Solve button: it is solved in closed form as soon as it appears.

Per-section gain

Control	Key	Default	What it does
Custom DC gain / HF gain / DC/HF gain	hw_dc_chk_*	off	Off: the section uses the gain Tab 3 gave it. On: type your own. The label follows where the passband is — DC for a low-pass, HF for a high-pass, DC/HF for a pure notch (equal at both ends). Greyed and forced on by Gained MFB and Eliminate R1 .
(gain value)	hw_dc_val_*	the allocated gain	2nd/3rd order: at least 0.1. Below 1 selects an attenuating cell where the family has one — a 2nd-order low-pass or any high-pass for VCVS; any section for MFB and AM. Moving gain between sections does not rebalance the others; keep the product equal to the passband gain.
Override section K_i value	hw_ki_chk_*	off	Band-pass sections, in place of the gain control. The passband of a band-pass is its peak, whose height depends on Q , so the coefficient K_i is set directly; the caption shows the pairing K_i and the resulting $ H(f_0) $.
K_i	hw_ki_val_*	the pairing K_i	In rad/s, or (rad/s) ² for a band-pass with an absorbed low-pass pole.
Realization	hw_fo_real_*	Non-inverting	1st-order sections: non-inverting (sign +1) or inverting (−1). Custom gain is then at least 0.01; non-inverting below 1 is an attenuator.

Solving and the candidates table

Sort by

Sens score

Descending

	#	Topology	Sens	Snap cost	DC gain	C1	C3	C4	R1	R2	R3	R5	R6
☐	0	3LP-gained	2.08	0.04	1.488	10n	82p	82p	7.87k	825k	1.15M	27.4k	56.2k
☐	1	3LP-gained	2.08	0.06	1.488	10n	68p	82p	9.09k	681k	1.47M	27.4k	56.2k
☐	2	3LP-gained	2.08	0.08	1.464	10n	120p	150p	6.81k	442k	953k	26.1k	56.2k
☐	3	3LP-gained	2.08	0.08	1.487	8.2n	120p	150p	8.66k	422k	953k	26.1k	53.6k
☐	4	3LP-gained	2.08	0.08	1.488	10n	150p	180p	8.66k	316k	681k	27.4k	56.2k
☐	5	3LP-gained	2.08	0.09	1.488	8.2n	390p	470p	10k	133k	261k	27.4k	56.2k
☐	6	3LP-gained	2.08	0.10	1.464	8.2n	180p	220p	8.66k	287k	619k	26.1k	56.2k
☐	7	3LP-gained	2.08	0.10	1.488	10n	150p	150p	7.87k	464k	619k	27.4k	56.2k
☐	8	3LP-gained	2.08	0.11	1.488	8.2n	120p	120p	9.53k	562k	787k	27.4k	56.2k
☐	9	3LP-gained	2.08	0.11	1.464	10n	180p	220p	7.15k	287k	619k	26.1k	56.2k

Pick a row to select that BOM.

Figure 8. Candidate BOMs for section 1, ranked by sensitivity. Component columns adapt to the cells that were solved.

Control	Key	Default	What it does
Solve section / Re-solve	hw_solve_*	—	Starts the solve in the background. Greyed while it runs, and when the chosen family has no cell for this section. Reads Re-solve when results for these exact settings exist.
Sort by	hw_sortf_*	Sens score	Sens score, Snap cost, the gain column, or any component column. Sorts by numeric value, so 8.25k comes before 78.7k.
Descending	hw_sortd_*	off	Reverses the order.
The table	hw_df_*	no row	Tick the box at the left of a row to pick that BOM.
Select solution #	hw_pick_*	0	Only on Streamlit versions without row selection: type the row's #.

Reading a row. # is the row's place after sorting; **Topology** the cell; **Sens** and **Snap cost** as in §2; the gain column (**DC gain**, **HF gain**, **DC/HF gain** or **Center gain**) is the passband gain the snapped parts actually give, op-amp included. Then one column per component the solved cells use. By default rows are ranked by Sens rounded to two decimals, with ties broken by the lower snap cost. A capacitor shown as 1n, 2n is two capacitors in parallel.

After a pick, a green box repeats the choice (*Selected #0: 3LP-gained · snap_cost 0.04 · sens 2.08 · DC gain 1.488 V/V*), the BOM is listed — capacitors as chosen, resistors snapped with the pre-snap ideal value in grey — and the schematic is drawn with designators prefixed by the section number (1R1, 1C1 ...) and the op-amp's name under U. With a real op-amp, a caption confirms *BOM checked against exact non-ideal op-amp physics*.

Control	Key	What it does
↓ SVG	*_svg_*	The annotated schematic as a vector file. Here and in Tab 5.
↓ PNG	*_png_*	The same as a bitmap. Only present when PNG export works (§6.8).

Overall filter

At the foot of the tab, once every section has a pick: **Overall realized DC gain** (HF gain for a high-pass) — the product of the sections' realized gains, signed — and **Overall realized passband gain**, which scales that by the design's passband-to-DC ratio. A band-pass shows only the centre gain; a band-reject shows **LF passband gain** and **HF passband gain** separately. A yellow box warns when the output is inverted (§6.5).

3.6 Tab 5 — Resulting Response & Schematic

Monte-Carlo tolerances perturbs every resistor and capacitor of the picked BOMs; op-amp parameters stay fixed.

Control	Key	Default	What it does
R max (kΩ)	<code>rtol_max_*</code>	blank = ∞	Resistor tolerance bands, $R_{\min} \leq R < R_{\max}$. Typing a maximum opens the next band, which starts there.
tol %	<code>rtol_tol_*</code>	1.00	Tolerance of the resistors in that band.
✕ remove last band	<code>rtol_rm</code>	—	Removes the last band.
Capacitor tol (%)	<code>resp_ctol</code>	5.00	Tolerance of every capacitor.
Runs	<code>resp_runs</code>	2000	10–20000. Above 1000 a caption warns it may take a few seconds.
Distribution	<code>resp_dist</code>	Gaussian (tol = 3σ)	Gaussian: the tolerance is three standard deviations. Uniform: equally likely anywhere within $\pm\text{tol}$.
Seed	<code>resp_seed</code>	0	The same seed and inputs give the same band — reruns are reproducible.
Envelope	<code>resp_envelope</code>	p1–p99	Which percentiles the shaded band spans: p1–p99, p5–p95 or min–max.
Run Monte-Carlo	<code>resp_mc_run</code>	—	Runs it and draws the band; a caption gives the DC-gain spread.
Linear Mag.	<code>resp_show_linear</code>	off	Linear magnitude axis.
Phase	<code>resp_show_phase</code>	off	Adds the phase.
Group delay	<code>resp_show_gd</code>	off	Adds the group delay, design and realized.

The plot, **Cascade response — design vs realized**, draws Ideal (design) in blue, Realized (BOM) in red, the Monte-Carlo band in grey with its median dashed, and a vertical line at each section's f_0 (and f_z). Below it, every section's schematic is stacked in cascade order with its own download buttons.

Report options

Generate Report

A4 PDF summary: specification, transfer function, pole-zero map with section pairing, per-section schematic + BOM, and the design-vs-realized Bode page.

☒ Cover page
 ☒ Per-section quality metrics ⓘ
 ☒ Monte-Carlo band + parameters

☒ Polynomial coefficient table
 ☐ Design warnings ⓘ
 ☒ Passband detail plot ⓘ

☒ Normalized prototype roots
 ☒ Phase & group delay on Bode page

Report title

Filter Synthesis Report

Project

Author

 Generate Report

Figure 9. The Generate Report block. Everything is on by default; Design warnings is greyed when there are none.

Control	Key	Default	What it does
Cover page	rep2_cover	on	A title page. The three boxes below appear only while it is ticked; unticked, the report uses the default title.
Report title	rep2_title	Filter Synthesis Report	Heading of page 1 and every footer.
Project	rep2_project	blank	Big title on the cover (blank: <i>Active Filter Design</i>).
Author	rep2_author	blank	On the cover.
Polynomial coefficient table	rep2_coeff	on	Report §2a/2b.
Normalized prototype roots	rep2_norm	on	Report §4a.
Per-section quality metrics	rep2_metrics	on	Sensitivity, snap cost, realized against target gain, per section.
Design warnings	rep2_warn	on	Carries the HF-resonance notes (§5) into the report. Greyed when none were raised.
Phase & group delay on Bode page	rep2_phgd	on	Adds them to report §6.
Monte-Carlo band + parameters	rep2_mc	on	The band, its settings and the passband-gain spread. If Monte-Carlo has not run with the current inputs, generating runs it once.
Passband detail plot	rep2_zoom	on	Report §7: the band of interest only. Stopband detail for a band-reject.
 Generate Report	report_btn	—	Builds the PDF. Greyed variants: <code>report_btn_blocked</code> (a section has no pick), <code>report_btn_nodata</code> and <code>report_btn_nobackend</code> (§6.5).
↓ Download PDF	report_d1	—	Saves <i>FilterReport_<response><type>_n<order><date>_<time>.pdf</i> .

4. Choosing a topology family

Each section chooses its own family, so one cascade can mix them.

	VCVS (Sallen-Key)	MFB (Friend)	AM (Ackerberg–Mossberg)
Op-amps per section	1	1	3
Polarity	Non-inverting; the pure notch inverts	Inverting; the notch cells do not	Inverting
Low-pass, high-pass, with or without notch	Yes	Yes	Yes
2nd-order band-pass, pure notch	Yes	Yes	Yes
3rd-order band-pass (absorbed real pole)	No	Yes	No
Section gain below 1	2nd-order low-pass; any high-pass	Any section	Any section
Strength	Fewest parts	Covers every section kind; gain is a free ratio	Q barely moves with op-amp bandwidth
Cost	Q sensitivity grows with Q	Inverts; Q-enhanced twins trade sensitivity for spread	Three op-amps; needs a low-output-resistance op-amp

MFB solves each cell together with its Q-enhanced (-QE) positive-feedback twin and ranks them; the plain cell usually wins where Q is modest. **AM** gets its accuracy from the matched pair $R_7 = R_8$, which compensates the op-amp's finite bandwidth: its Q error grows with $(f_o/f_t)^2$ instead of $Q \cdot f_o/f_t$.

Rules of thumb:

- **Start with VCVS.** It is the cheapest realization of low-pass and high-pass sections.
- **A 3rd-order band-pass section needs MFB** — or untick absorption in Tab 3 so the real pole becomes its own 1st-order section.
- **A section gain below 1** on a 3rd-order low-pass needs MFB or AM, or move that gain to another section in Tab 3.
- **A real op-amp drags the realized curve away near a high-Q f_o :** try AM with a low- R_o op-amp.
- **Polarity matters:** count the inverting sections; the Overall filter readout warns when the total is odd.
- **No BOM at all:** before changing family, read the diagnosis (§6.4) — it often says the family is fine and the envelope is not.

5. Op-amps and what they do to the design

The op-amp is chosen per section, because a high-Q or high-gain section demands more bandwidth than the others.

Library entry	A _{ol} (V/V)	GBWP	R _o
Ideal (no op-amp limits)	∞	∞	0
AD8505 / AD8506 / AD8508	$1 \cdot 10^5$	95 kHz	1 k Ω
LMV358A	$1 \cdot 10^5$	1 MHz	1.2 k Ω
MAX9636/MAX9637/ MAX9638	$1 \cdot 10^5$	1.5 MHz	100 Ω
MAX40100	$1.41 \cdot 10^6$	1.5 MHz	100 Ω
Custom...	your values	your values	your values

- **A_{ol}** — finite DC gain: small gain and Q errors.
- **GBWP** — the op-amp's gain falls with frequency. This shifts f_0 and raises Q, more so for high-Q, high-gain sections and as f_0 approaches the GBWP.
- **R_o** — output resistance works against the feedback network and shows up far above the passband, as a rise of the realized response.

With a real op-amp the solver pre-distorts the ideal solution so that the section lands on target *with* that op-amp, then snaps and checks the BOM against the exact non-ideal model. Tab 5's realized curve and the report use the same model; Monte-Carlo holds it fixed. The part number (or the Custom parameters) is printed under the op-amp in the schematic.

The HF resonance warning. Tab 5 compares realized with design from the highest corner up to 100 times it. When red rises at least 1 dB above blue, a yellow box appears:

⚠ HF resonance (~N dB at F) — over [range] the realized (red) response peaks ~N dB above the design (blue). This is the Ackerberg–Mossberg finite-R_o loop resonance from the op-amp output impedance (AM section ...), not a synthesis error.

Without AM sections it reads ⚠ **HF rise** and blames the output impedance alone. It is physics, not a solver fault. Fix it with an op-amp of lower output resistance (about 50 Ω or less), or with MFB for the sections it names.

6. When it goes wrong

Messages are quoted as the app prints them; "..." marks the parts that vary.

6.1 Specification and approximation

The app says	What it means	What to do
Mathematical Constraint Violation: Band-Reject filters require an EVEN total order. Your current total is N.	Asymmetric band-reject with LP + HP odd.	Change one of the two orders by one.
Mathematical Constraint Violation: Asymmetric Elliptic Band-Reject requires ...	Asymmetric elliptic band-reject needs even orders that differ by at most 2.	Adjust the orders.
 Max total order for Asymmetric Elliptic is 15. Extreme orders may cause calculation delays.	A warning, not a block.	Lower LP + HP, or be patient.
Engine Error: ...	The approximation engine failed for this specification.	Change the specification slightly (order, corners, A_s). If it persists, report it (§6.7).
Stopband Attenuation does not meet the requirements (Asymptote exceeds limit)	The stopband never reaches A_s — typically after pinning notches.	Unpin or move notches, raise the order, or lower A_s.
Calculated Stopband Edge (f_s): ... (Extended due to manual notch placement)	A pinned notch pushed the stopband edge out.	Informational. Move the notch if f_s matters.
Order is too low to support finite transmission zeros.	No notch rows at this order.	—
Manual Notch Tuning ... is disabled ...	Elliptic band filters, and Elliptic above order 8.	By design.

6.2 Pairing

The app says	What it means	What to do
 Pairing Incomplete: There are floating zeros on the board (drawn in gray). Please assign them to a stage.	A manual edit left zeros without a section; nothing is handed to the Topology tab.	Click a pole, then the grey zero — or press  Auto-Pair (Reset).
 Please click a Pole first to initiate routing.	A zero was clicked first.	Click a pole, then the zero.
 Cannot merge poles: 'Enable 3rd-Order Sections' is unchecked.	Absorbing a real pole needs the checkbox.	Tick it.
 Manual Override Active: Background auto-router is currently bypassed.	Your edits are in charge.	 Auto-Pair (Reset) to go back.

6.3 Topology, before and during a solve

The app says	What it means	What to do
Finalize the cascade in Biquad Pairing & Cascading first — each section will appear here for hardware synthesis.	No cascade yet.	Complete the pairing in Tab 3 (§6.2), or fix the specification.
 ... section (family ...) — solver not yet available; gated to avoid mis-solving on a mismatched cell.	No solver exists for this kind of section yet.	Re-pair so the section becomes a supported kind — e.g. untick absorption.
Section N is a 3rd-order band-pass ... Switch this section to MFB (Friend) ...	VCVS and AM cannot place the third pole of a band-pass. Solve is greyed.	Family → MFB, or give the real pole its own section.
No R series selected — defaulting to E48.	No resistor series ticked.	Tick at least one.
 E3 must be added to <code>unified_solver_v2._E_SERIES</code> , else the cap grid is empty.	E3 capacitors are not enabled in the solver yet.	Use E6 or E12.
 Solving... (other sections and tabs stay responsive)	Normal.	Keep working elsewhere.

6.4 A solve that returns nothing

When a solve finds no BOM, the app runs a quick idealized check of what the section would need and replaces the generic message with a diagnosis. The diagnosis can take up to half a minute after the solve ends.

Section 1 · order 3 · $f_p = 2,000 \text{ Hz}$ · $f_o = 2,000 \text{ Hz}$ · $Q = 0.618$

>  Section 1 — topology & component settings

Cell: 3LP-gained · $H(0) = 1.5$ (from K) · tol $\pm 2\%$ ☐ Custom DC gain Re-solve

A realizable design exists, but not inside your envelope — it needs a resistor ratio $\approx 539\times$ (your Max R ratio is $5\times$); a capacitor ratio $\approx 517\times$ (your $C_{\text{max}}/C_{\text{min}}$ is $1\times$). Widen those limits (or add more R series) and re-solve.

Figure 10. The envelope is too tight: the message names the resistor and capacitor ratios the design needs, and the limits you set.

The app says	What it means	What to do
No realization inside the component envelope. Widen R/C limits, relax Max R ratio, add more R series, or raise thoroughness.	Nothing found, and the check reached no verdict.	In that order.
A realizable design exists, but not inside your envelope — it needs a resistor ratio $\approx N\times$ (your Max R ratio is $M\times$); a capacitor ratio $\approx \dots\times$ (your $C_{\text{max}}/C_{\text{min}}$ is $\dots\times$). ...	The design needs a wider spread than you allow; each part names the limit that stops it.	Raise Max R ratio above N ; widen $C_{\text{min}}\dots C_{\text{max}}$ beyond the capacitor ratio named.

The app says	What it means	What to do
A realizable design exists inside your envelope (it needs only ...), but the search didn't reach it. ...	The limits are fine; the search missed it.	Search thoroughness → Thorough and re-solve, or nudge the section gain slightly.
A realizable design almost certainly exists at this gain (...), but the search didn't find one in your envelope. ...	The same, found by a gain-band check.	The same.
No realizable solution at this gain. You asked for X, but the nearest gain this topology can realize for the target shape is $\approx Y$	This family cannot make this gain with this f_0 and Q (and f_z).	Set the section gain to about Y with Custom gain and move the difference to another section, or change the family.
No realizable solution exists for these targets — the pole/zero shape itself isn't achievable with this topology. Choose another topology.	Structural.	Change the family (§4).
No realization fits the constraints — raise $C_{\max}$, add a resistor E-series, or (ni-gained) relax the gain so $R3+R4$ lands in 5k–50k.	A 1st-order section found no parts.	As it says.

6.5 Results and the report

Resulting Response & Schematic

Pick a BOM for every section in the **Topology** tab to see the cascade response (still pending: section 1, 2).

Generate Report

⚠ Report generation is not available yet — section 1, 2 still have no selected BOM. Solve and pick a row for every section in the **Topology** tab.

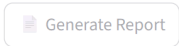
 Generate Report

Figure 11. Solved is not picked: the report stays blocked and names the sections without a BOM.

The app says	What it means	What to do
Select a solution for every section to see the overall gain (still pending: section N).	Tab 4's Overall filter waits for picks.	Tick a row in each section listed.
⚠ Output is inverted — an odd number of inverting (inv) stages gives overall sign -1	The cascade inverts.	Make the count of inverting sections even (family or Realization), or add an inverter.
Finalize the cascade and synthesize sections in the Topology tab first — this view combines the selected per-section BOMs.	No cascade yet.	Tabs 3 and 4 first.
Pick a BOM for every section in the Topology tab to see the cascade response (still pending: section N).	Solved is not picked.	Tick a row in each section listed.

The app says	What it means	What to do
⚠ Report generation is not available yet — section N still has no selected BOM. ...	The same gate, for the report.	The same.
Section N: can't build response for cell (...).	The response model of that cell failed.	Pick another row; report it (§6.7).
⚠ HF resonance (~N dB at F) ... / ⚠ HF rise (~N dB at F) ...	Op-amp output resistance (§5).	A lower- R_o op-amp, or MFB for the AM sections named.
Inputs changed since the last run — click Run Monte-Carlo to refresh the band.	The band on screen is stale.	Run Monte-Carlo again.
Monte-Carlo failed: ...		Report it.
Monte-Carlo has not been run (or its inputs changed). Generating the report will run it once with the settings above — this adds a few seconds.	Informational.	—
Settings changed since this PDF was built — press Generate Report again to refresh it.	The download button still offers the older PDF.	Generate again.
Report generation failed: ...		Report it.
PDF report support needs the reportlab package, which is not installed in this environment.	Optional packages missing (source install).	<code>pip install reportlab matplotlib cairosvg</code>
Report data is not being published by app.py — missing ...	The installation is incomplete or modified.	Reinstall.
⚠ Schematic SVG not found ... expected ... in ...	The drawing for that cell is missing.	Restore the Section_Schematic_Diagrams folder (§7.4).

6.6 Errors with a Details box

Engine Error: ... (under the title) and **Solver error:** ... (in a section) each come with a collapsed **Details (paste this into a bug report)** box holding the technical trace. The first is the approximation engine, the second the hardware solver. Neither is your fault; report both (§6.7).

6.7 Reporting a problem

Include:

- the app version from the title line (*Filter Synthesizer v1.0.3 - ...*);
- the specification — the sidebar, or page 2 of a PDF report;
- the section's header line (order, fp, f_0 , Q, f_z) and a screenshot of its open settings expander;
- the text of the **Details** box, when there is one;

- for the packaged app, the output of `FilterSynthesizer.exe --selftest` , which writes a diagnostic report without starting the app.

A maintainer may ask you to start the app with the environment variable `FILTERSYNTHESIZER_DEBUG=1` ; each section then shows a  **exact solver call** panel whose contents belong in the report.

6.8 No ↓ PNG button next to ↓ SVG

The PNG button appears only when the `cairosvg` package can render — it is a live indicator. Without it you lose the PNG downloads, and the report's schematic pages print a placeholder; everything else works. From source: `pip install cairosvg` (on Windows it also needs the Cairo library). In the packaged app, use the SVG.

7. Reference tables

7.1 E-series

Series	Values per decade	Usual tolerance class	Offered for
E3	3	wider than $\pm 20\%$	capacitors (not usable yet)
E6	6	$\pm 20\%$	capacitors
E12	12	$\pm 10\%$	capacitors, resistors
E24	24	$\pm 5\%$	resistors
E48	48	$\pm 2\%$	resistors
E96	96	$\pm 1\%$	resistors

The series decides which values the solver may use. The tolerance of the parts you actually buy goes into Tab 5's Monte-Carlo.

7.2 Defaults

Setting	Default	In real parts
C_min ... C_max	6.80e-05 ... 1.00e-02 μF	68 pF ... 10 nF
R_min ... R_max	0.3 ... 2000 k Ω	300 Ω ... 2 M Ω
Max R ratio	500	
Capacitor / resistor series	E12 / E48	
Family / op-amp	VCVS / Ideal	
Monte-Carlo	R 1 %, C 5 %, Gaussian, 2000 runs, seed 0, p1–p99	

7.3 Reading a cell name

Part	Means	Examples
Leading digit	Section order	2LP-unity , 3LP-gained
LP, HP, BP, N	Low-pass, high-pass, band-pass, pure notch	2HP-gained , 2BP , 2N
n after LP or HP	With a notch (a zero pair)	2LPn-unity , 3HPn-MFB
2BP1LP, 2BP1HP	Band-pass pair plus an absorbed low- or high-pass pole: a 3rd-order band-pass	2BP1HP-MFB
-unity, -gained, - atten	VCVS gain variant: = 1, > 1, < 1	2LP-atten
-MFB, -AM	Family; no suffix means VCVS	2LP-MFB , 3LP-AM

Part	Means	Examples
-MFB2, -AM2	The family's second realization: MFB2 the unity/gained HP-notch, AM2 the alternative output tap	2HPn-MFB2 , 2LP-AM2
-QE	Q-enhanced twin (positive feedback)	2BP-MFB-QE
-LS	Lower-sensitivity branch	2LPn-MFB-LS
+R1, +R7, +R8	Twin with one extra resistor	2LPn-MFB-LS+R7
-C1s	Input capacitor split into a parallel pair	2HP-AM-C1s
1LP-ni, 1LP-inv	1st order, non-inverting / inverting, then the gain variant	1LP-ni-gained

7.4 Files and folders

Where	What
The folder holding FilterSynthesizer.exe	<code>_internal</code> (must stay next to the EXE) and <code>Section_Schematic_Diagrams</code> — the schematic drawings, one <code>.drawio.svg</code> per cell. A drawing dropped there is used without a rebuild.
<code>%LOCALAPPDATA%\FilterSynthesizer</code> (Windows); <code>\$XDG_DATA_HOME/FilterSynthesizer</code> or <code>~/.local/share/FilterSynthesizer</code> (Linux)	The symbolic transfer-function cache and logs. Safe to delete; the next start is slow again (10–20 s instead of 2–4 s). Running from source, the cache sits in the app folder.
Your browser's download folder	Reports, SVG and PNG schematics.

7.5 Optional packages

Package	Without it
reportlab, matplotlib	No PDF report. The report block names the missing package and the pip line.
cairosvg	No ↓ PNG schematics; the report's schematic pages print a placeholder.

All three are in `requirements.txt` ; the usual casualty is cairosvg, which needs the Cairo library on Windows.

8. What the PDF report contains

A4, the plot pages in landscape. Every page carries a footer: report title · date and time · design hash · page. The design hash identifies the cascade — two reports with the same hash describe the same sections and gains.

Part	Contents	Controlled by
Cover	Project (or <i>Active Filter Design</i>), the subtitle <i>Response Type</i> · order <i>N</i> · corner, author, date, app version	Cover page
1 · Design specification	Every sidebar value, and the calculated stopband edge(s)	always
2 · Transfer function	$H(s)$, denormalized in rad/s, with the gain constant K ; 2a monic coefficients; 2b coefficients with K folded into the numerator	2a/2b: Polynomial coefficient table
3 · Pole–zero map — section pairing	Each section's roots in its colour; dashed links pair poles with zeros; dotted circles mark each section's ω_0	always
4 · Root locations	Poles and zeros in rad/s with their section; 4a the normalized roots	4a: Normalized prototype roots
5.N · Section N — cell	Design parameters (order, f_0 , ω_0 , Q , f_z , f_p , K_i , Peak $ H $); schematic; BOM with the pre-snap ideal in brackets; op-amp; solver constraints; quality metrics; the section's $H(s)$	Quality metrics: Per-section quality metrics
6 · Bode — design vs realized	Cascade magnitude with the Monte-Carlo band and its parameters; the passband-gain table	Phase & group delay on Bode page; Monte-Carlo band + parameters
7 · Passband detail	Design vs realized over the band of interest, same shading (<i>Stopband detail</i> for a band-reject)	Passband detail plot
Design warnings	The HF-resonance notes, when any were raised	Design warnings

Gain error in the quality metrics. With a single section it is measured against the specified filter gain. In a cascade each section carries only its share, so it is measured against that section's own design target. *DC gain (design)* is evaluated at the section's passband reference — f_0 for a band-pass, DC for a low-pass, HF for a high-pass; on a 3rd-order section the absorbed pole tilts the response, so it sits slightly below Peak $|H|$. *DC gain (realized)* uses the snapped values at zero tolerance.

Passband gain in §6 is the top of the passband ripple — the highest magnitude inside the band — found separately for the design, the realized curve and every Monte-Carlo run. It is not the DC gain: for an even-order Chebyshev or Elliptic without the even-order modification, DC sits a full ripple lower. This is also why the report's spread can exceed the DC-gain spread shown in Tab 5. In the Quick Start example the screen gives 3.14 ... 3.75 dB at DC, while the report's highest per-run peak is +5.07 dB: tolerances raise a peak below the corner, and the report's table shows the peaks landing anywhere from 200 Hz to 1.53 kHz.